Finite terms (checked: red = remove, blue = add)

Key. Red: remove. Blue: add. The checks and the derivations are in `Finite_terms_final.pdf`. Every row not marked is correct as written: it equals the row of `Finite_terms_1.tex` minus its BFKL ($\log \frac{\vee}{k^+}$) and JIMWLK ($\log \frac{\vee}{\Lambda}$) terms, term by term. The changes are:

C1 Group II Row IV: the line $-\log \frac{\vee}{k^+} H_4$ is missing. H_4 is in the BFKL list, and the ξ integral of line 1 also contains it.

C2 Group III Rows III (both parts), V, VI: $\left[\frac{1}{p^+}\right]_+ \to \left[\frac{1}{p^+}\right]_+^{(k^+)}$, where $\int_0^{\vee - k^+} dp^+ \left[\frac{1}{p^+}\right]_+^{(k^+)} F(p^+) \equiv \int_0^{\vee - k^+} dp^+ \frac{F(p^+) - F(0)\frac{k^+}{p^+ + k^+}}{p^+}$.

C3 Group III Rows III (both parts), V, VI: subtract the terms (B) (the region $y' \to z$). They are in the BFKL list.

C4 Group I Row III: the first term of d^3N_{3a} and the last term of d^3N_{3b} (edge kernel $\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}$) are removed. The edge kernel is replaced by $\mathcal{D}$ (fix (A)). The last term of d^3N_{3b} is not integrable at $z \to w$.

C5 Group III Row III part 2: the measure is $\int_{w,x',y',z,x}$.

The DGLAP and running-coupling terms sit in Group I Row II part 1, Group I Row V 2b UV and Group II Row IV (notes under each).

1 Group I (with c.c.)

1.1 Row I

NA

1.2 Row II

$$\begin{aligned}\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[\left[-\frac{2}{\epsilon}-2\gamma-\log\left[\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\times\right.\\ \left.\left[\frac{11}{3}+\log\left[1-\frac{k^+}{\vee}\right]\right]-\frac{67}{9}+\frac{\pi^2}{3}+\log^2\left[1-\frac{k^+}{\vee}\right]+\log\left[1-\frac{k^+}{\vee}\right]\log\left[\frac{\vee}{k^+}\right]\right]\times\\ \Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\Big]\end{aligned} \tag{1.1}$$

No change. In the final result: the $\frac{11}{3}$ term is running coupling and cancels against $-\frac{11}{6}$ of Row V 2b UV. The constant $-\frac{67}{9}+\frac{\pi^2}{3}$ gives $\frac{\alpha_s}{2\pi}N_c\left(\frac{67}{9}-\frac{\pi^2}{3}\right)$LO. The pole bracket times $\log(1-\frac{k^+}{\vee})$ is Q1. $\log^2+\log\log$ is $V_\vee$, a finite term. (`Finite_terms_final.pdf`, Secs. 3, 4.)

$$\begin{aligned}\frac{d^3N_{2\,b}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4}{64\pi^6}\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int d^2p\,d^2k\int_{x,y} e^{ik\cdot(w-y)}\,e^{ip\cdot(y-x)}\,\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{k}^2}\,R^i(\mathbf{k},\mathbf{p})\\ \times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{hbc}\,U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}-f^{abc}\,U^{bd}(x)\,U^{ce}(y)\,\{\rho^d(x),\rho^e(y)\}\Big]\end{aligned} \tag{1.2}$$

where

$$R^i = -2k^i\log\left(\frac{p^2}{k^2+\frac{k^+}{\vee}p^2}\right)+\frac{(2p^2-2\,k\cdot p+k^2)\,k^i+k^2p^i}{(k-p)^2}\log\left[\frac{k^2}{(k-p)^2}+\frac{k^+}{\vee}\left(1-\frac{k^2}{(k-p)^2}\right)\right]-\frac{p^2k^i+k^2p^i}{(k-p)^2}\log\left(1-\frac{k^+}{\vee}\right). \tag{1.3}$$

No change.

1.3 Row III

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)} \Bigg\{ \\
&- \frac{ig^4\, f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
&+ \frac{ig^4\, f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left[\frac{\left[(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2\right]}{\left[\left(1+\frac{\Lambda}{k^+}\right)(x-z)^2-(x-w)^2\right]}\right] \\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
&- \frac{ig^4\, f^{abd}}{16\pi^5}\,\frac{1}{k^+}\left(-\log\left[1-\frac{\Lambda}{k^+}\right]\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
&+ \frac{ig^4\, f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left[\frac{\left[(k^+-\Lambda)\,(x-z)^2-k^+\,(x-w)^2\right]}{\left[\Lambda\,(x-z)^2-k^+\,(x-w)^2\right]}\right] \\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
&- \frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\,\log\left[\frac{\left[\frac{k^+}{\vee}\,(x-w)^2-(x-z)^2\right]}{\left[(x-w)^2-\frac{\Lambda}{k^+}\,(x-z)^2\right]}\right] \\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \qquad (1.4)
\end{aligned}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)} \Bigg\{ \\
&- \frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,(x-w)^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
&+ \frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
&- \frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
&+ \frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
&- \frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-w)^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
&+ \frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
&- \frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-z)^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]
\end{aligned}
$$

$$
+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}
$$

$$
\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]
$$

$$
-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}
$$

$$
\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\Bigg\} \tag{1.5}
$$

$$
\frac{d^3N_{3a\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\int_{w,w'}e^{-ik(w'-w)}\Bigg\{
$$

$$
-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}
$$

$$
\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]
$$

$$
-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}
$$

$$
\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]
$$

$$
+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}
$$

$$
\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]
$$

$$
-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}
$$

$$
\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]
$$

$$
+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}
$$

$$
\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]
$$

$$
-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}
$$

$$
\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]
$$

$$
+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right](x-z)^2}
$$

$$
\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]
$$

$$
+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right](x-w)^2}
$$

$$
\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]
$$

$$
-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}
$$

$$
\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]
$$

$$
+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2}
$$

$$
\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]
$$

$$
+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2}
$$

$$
\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]
$$

$$
-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'}\frac{(x-w)\cdot(x-z)\,(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2}
$$

$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \Bigg\} \tag{1.6}$$

$$\begin{aligned}
\frac{d^3N_{3b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)} \Bigg\{ & \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'} \frac{(x-w)\cdot(x-z)\,(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
& -\frac{ig^4 f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\left[k^+(x-w)-(k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w)-\Lambda(x-z)\right]^2}\right]\int_{x,y,z,x'} \frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
& +\frac{ig^4 f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\left[k^+(x-w)-(k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w)-\Lambda(x-z)\right]^2}\right]\int_{x,y,z,x'} \frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
& -\frac{ig^4 f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\left[k^+(x-w)-(k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w)-\Lambda(x-z)\right]^2}\right]\int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2\,(z-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \Bigg\}
\end{aligned} \tag{1.7}$$

where

$$\begin{aligned}
\mathcal{A} = \frac{1}{\sqrt{(x-w)^2(x-z)^2 - [(x-w)\cdot(x-z)]^2}} \Bigg\{ & \arctan\left[\frac{(k^+-\Lambda)(x-z)^2 - k^+(x-w)\cdot(x-z)}{k^+\sqrt{(x-w)^2(x-z)^2 - [(x-w)\cdot(x-z)]^2}}\right] \\
& - \arctan\left[\frac{\Lambda(x-z)^2 - k^+(x-w)\cdot(x-z)}{k^+\sqrt{(x-w)^2(x-z)^2 - [(x-w)\cdot(x-z)]^2}}\right] \Bigg\}
\end{aligned}$$

Change C4. The two red terms have the edge kernel $\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}$. Their $\log\frac{\vee}{k^+}$ and $\log\frac{\vee}{\Lambda}$ partners are replaced by $\mathcal{D}$ (fix (A), `BFKL_final.pdf`; fix F3 of `Evolution_terms.tex`). The edge kernel therefore does not belong to the row, and its finite terms go with it.

- The first term of d^3N_{3a} vanishes for $\Lambda \to 0$.
- The last term of d^3N_{3b} is, for $\Lambda \to 0$, $\log\frac{(z-w)^2}{(x-w)^2}$ times the edge kernel. Near $z \to w$ it behaves like $\frac{\log(1/|z-w|)}{(z-w)^2}$, so $\int_z$ diverges. Numerically: the angular average of the row behaves like $\frac{\log(1/r)}{r^2}$, $r = |z-w|$; without this term it is integrable (`edge_check.json`).
- In its place belongs the finite part of the $\mathcal{D}$ region. It needs the p^+ integral of this region with the kernel of $\mathcal{D}$. That integral is not in your notes or in mine (open, Q3).

All other terms of Row III are finite. On the circle $(x-z)^2 = (x-w)^2$ the $\frac{1}{[(x-z)^2-(x-w)^2]^n}$ cancel among them (`edge_check.json`).

1.4 Row IV

$$\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)} \Bigg\{ & \\
& \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2} \times \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2 + \frac{k^+}{\vee}(x-w)^2}{\frac{\Lambda}{k^+}(x-z)^2 + (x-w)^2}\right] \times \left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right. \\
& \left. -\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2} + \frac{\delta^{ij}}{2\,(z-w)^2}\right] \\
& -\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2} \times \left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right) \\
& +\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2} \times \log\left[1-\frac{k^+}{\vee}\right]\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} + \frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right] \Bigg\} \times \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{e}(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{e}(y)\rho^{d}(x)\Big]
\end{aligned} \tag{1.8}$$

1.5 Row V

First part

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^{+\,2}}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\\
&\left[\delta_{im}\,\delta_{jk}-\left[\frac{k^+}{p^+}\right]_+\delta_{jm}\,\delta_{ik}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\\
&\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dq^+}{2\pi}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\,\left[\frac{1}{q^+}\right]_+\\
&\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]\\
&-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-z)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{\Lambda}{k^+}\right]\,\frac{(x'-w')\cdot(y-z)\,(z-x)\cdot(y'-x)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]\\
&-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-x)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{\Lambda}{k^+}\right]\,\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\\
&\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]
\end{aligned}
\tag{1.9}
$$

Second part

$$
\begin{aligned}
\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}\right.\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\int_{k^+}^0\frac{dq^+}{2\pi}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{1}{[(k^+-q^+)\,(y-x)^2+q^+\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\left\{\left[\frac{k^+}{q^+}\right]_+\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]-\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right\}\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-z)^2}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}\right.\\
&\left.+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-x)^2}
\end{aligned}
$$

$$\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \tag{1.10}$$

$$\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\left[2\log\left(1-\frac{\Lambda}{k^+}\right)-\frac{11}{6}\right]\int_{y,z,x',w'}e^{-ik\cdot(w'-z)}$$
$$\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[U^{af}(y)-U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y). \tag{1.11}$$

No change. In the final result: $-\frac{11}{6}$ cancels the $\frac{11}{3}$ of Row II part 1 (running coupling). $2\log(1-\frac{\Lambda}{k^+})\to 0$. Nothing of this term is left among the finite terms.

$$\begin{aligned}
\frac{d^3N_{2b\,\mathrm{finite\,NLO}}}{d^3k}=&\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\times\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}}\right.\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\right]\\
&\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\left[\frac{k^+}{q^+}\right]_+\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+-\Lambda}^\Lambda\frac{dq^+}{2\pi}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x',w'}e^{-ik\cdot(w'-x)}\left[\frac{(y-x)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right.\\
&\left.-\frac{(y-x)\cdot(x'-w')\,(y-z)\cdot(z-x)}{2(y-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\left[\frac{(y-z)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right.\\
&\left.+\frac{(y-x)\cdot(z-x)\,(y-z)\cdot(x'-w')}{2(y-x)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\left[-\frac{11}{6}+2\log\left[1-\frac{\Lambda}{k^+}\right]\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned} \tag{1.12}$$

2 Group II

2.1 Row I

2.1.1 First part

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik(w'-w)}\Big\{$$

$$
\begin{aligned}
&- \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]\times\\
&\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]\\
&+\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\\
&-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\\
&\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]\\
&+\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik(w'-w)}\Bigg\{\\
&\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w')^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w')^2}\right]\times\\
&\left[\frac{\delta^{ij}}{2\,(z-w')^2}+\frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2}-\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right]\\
&-\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\,\frac{(x'-w)\cdot(y-z)}{(x'-w)^2\,(y-z)^2\,(z-w')^2}\\
&+\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right]\Bigg\}\\
&\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right]
\end{aligned}
\tag{2.1}
$$

2.1.2 Second part

$$
\begin{aligned}
\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k}&=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\Bigg\{\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\times\\
&\left[\frac{(x-w)\cdot(z-w')\,(x'-z)\cdot(z-w)}{(z-w)^2\,(z-w')^2}-\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(x'-z)}{2\,(z-w)^2\,(x'-w')^2}\right.\\
&+\frac{(x-z)\cdot(x'-z)\,(x-w)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-w')^2}\\
&+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(x'-z)}{2\,(z-w)^2\,(x'-z)^2}\\
&\left.-\frac{(x-z)\cdot(z-w')\,(x'-w')\cdot(x-w)}{2\,(x-w)^2\,(z-w')^2}-\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(x-z)}{4\,(x-w)^2\,(x'-z)^2}\right]\times\\
&\left[\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{k^+}{\vee}\,(x'-w')^2+\left(1-\frac{k^+}{\vee}\right)(x'-z)^2}\right]-\log\left[\frac{\Lambda\,(x-z)^2+k^+\,(x-w)^2}{k^+\,(x'-w')^2+\Lambda\,(x'-z)^2}\right]\right]\Bigg\}\times\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\right]
\end{aligned}
\tag{2.2}
$$

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k}&=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times\\
&\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.\\
&\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\left[(x-z)^2-\frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\right]\times
\end{aligned}
$$

$$
\Bigg[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x') \rho^{e}(x)
$$

$$
- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Bigg]
$$

$$
- \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x-z)^2}{\left[(x'-w')^2\,(x-z)^2\,(x-w)^2 - (x-w)^4\,(x'-z)^2\right]} \times
$$

$$
\Bigg[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} + \frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}
$$

$$
+ \frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2} + \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2} \Bigg]
$$

$$
\times \log \left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2 + \frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2 + (x-w)^2} \right] \times
$$

$$
\Bigg[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x') \rho^{e}(x)
$$

$$
- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Bigg]
$$

$$
+ \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-z)^2}{\left[(x'-w')^4\,(x-z)^2\, - (x-w)^2\,(x'-z)^2\,(x'-w')^2\right]} \times
$$

$$
\Bigg[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} + \frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}
$$

$$
+ \frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2} + \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2} \Bigg]
$$

$$
\times \log \left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2 + \frac{k^+}{\vee}\,(x'-w')^2}{\frac{\Lambda}{k^+}\,(x'-z)^2 + (x'-w')^2} \right]
$$

$$
\Bigg[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x') \rho^{e}(x)
$$

$$
- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Bigg] \tag{2.3}
$$

$$
\frac{d^3 N_{2c\,\mathrm{NLO}}}{d^3 k} = \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-w')^2}{(x'-z)^2}\, \frac{1}{\left[(x'-w')^2\,(x-z)^2 - (x-w)^2\,(x'-z)^2\right]}
$$

$$
\Bigg[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2} - \frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}
$$

$$
+ \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2} \Bigg] \times \log \left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2 + \frac{k^+}{\vee}\,(x'-w')^2}{\frac{\Lambda}{k^+}\,(x'-z)^2 + (x'-w')^2} \right]
$$

$$
\Bigg[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x') \rho^{e}(x)
$$

$$
- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Bigg]
$$

$$
- \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x-w)^2}{(x-z)^2}\, \frac{1}{\left[(x'-w')^2\,(x-z)^2 - (x-w)^2\,(x'-z)^2\right]}
$$

$$
\Bigg[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2} - \frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}
$$

$$
+ \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2} \Bigg] \times \log \left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2 + \frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2 + (x-w)^2} \right]
$$

$$\Bigg[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x') \rho^{e}(x)$$

$$- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Bigg] \tag{2.4}$$

$$\frac{d^3 N_{2d\,\mathrm{NLO}}}{d^3 k} = -\frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+}\, \left(-\log\left[1 + \frac{\Lambda}{k^+} \right] \right) \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{1}{[(x-z)^2 - (x-w)^2]}\, \frac{1}{[(x'-w')^2 - (x'-z)^2]}$$

$$\Bigg[\frac{(x-w)\cdot(z-w)\, \left[(x'-z)^2 - (x'-w')^2 \right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x-w)\, \left[(x'-z)^2 - (x'-w')^2 \right]}{4\,(x-z)^2\,(z-w')^2}$$

$$+ \frac{(x'-w')\cdot(z-w')\, \left[(x-z)^2 - (x-w)^2 \right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\, \left[(x-z)^2 - (x-w)^2 \right]}{4\,(z-w)^2\,(x'-z)^2} \Bigg]$$

$$\Bigg[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x') \rho^{e}(x)$$

$$- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Bigg]$$

$$+ \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x-z)^2}{[(x-z)^2 - (x-w)^2]}\, \frac{1}{[(x'-w')^2\,(x-z)^2 - (x-w)^2\,(x'-z)^2]}$$

$$\Bigg[\frac{(x-w)\cdot(z-w)\, \left[(x'-z)^2 - (x'-w')^2 \right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x-w)\, \left[(x'-z)^2 - (x'-w')^2 \right]}{4\,(x-z)^2\,(z-w')^2}$$

$$+ \frac{(x'-w')\cdot(z-w')\, \left[(x-z)^2 - (x-w)^2 \right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\, \left[(x-z)^2 - (x-w)^2 \right]}{4\,(z-w)^2\,(x'-z)^2} \Bigg] \times$$

$$\log\left[\frac{\left(1 - \frac{k^+}{\vee} \right) (x-z)^2 + \frac{k^+}{\vee}\, (x-w)^2}{\frac{\Lambda}{k^+}\, (x-z)^2 + (x-w)^2} \right] \times$$

$$\Bigg[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x') \rho^{e}(x)$$

$$- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Bigg]$$

$$+ \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-z)^2}{[(x'-w')^2 - (x'-z)^2]}\, \frac{1}{[(x'-w')^2\,(x-z)^2 - (x-w)^2\,(x'-z)^2]}$$

$$\Bigg[\frac{(x-w)\cdot(z-w)\, \left[(x'-z)^2 - (x'-w')^2 \right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x-w)\, \left[(x'-z)^2 - (x'-w')^2 \right]}{4\,(x-z)^2\,(z-w')^2}$$

$$+ \frac{(x'-w')\cdot(z-w')\, \left[(x-z)^2 - (x-w)^2 \right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\, \left[(x-z)^2 - (x-w)^2 \right]}{4\,(z-w)^2\,(x'-z)^2} \Bigg] \times$$

$$\log\left[\frac{\left(1 - \frac{k^+}{\vee} \right) (x'-z)^2 + \frac{k^+}{\vee}\, (x'-w')^2}{\frac{\Lambda}{k^+}\, (x'-z)^2 + (x'-w')^2} \right] \times$$

$$\Bigg[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x') \rho^{e}(x)$$

$$- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Bigg] \tag{2.5}$$

$$\frac{d^3 N_{2e\,\mathrm{NLO}}}{d^3 k} = \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+}\, \left(-\log\left[1 + \frac{\Lambda}{k^+} \right] \right) \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{1}{[(x-z)^2 - (x-w)^2]}\, \frac{1}{[(x'-w')^2 - (x'-z)^2]}$$

$$\Bigg[\frac{(x-z)\cdot(z-w)\, \left[(x'-z)^2 - (x'-w')^2 \right]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x-z)\cdot(x-w)\, \left[(x'-z)^2 - (x'-w')^2 \right]}{4\,(x-w)^2\,(z-w')^2}$$

$$+ \frac{(x'-z)\cdot(z-w')\, \left[(x-z)^2 - (x-w)^2 \right]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x'-w')\, \left[(x-z)^2 - (x-w)^2 \right]}{4\,(z-w)^2\,(x'-w')^2} \Bigg]$$

$$\Bigg[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x') \rho^{e}(x)$$

$$
\begin{aligned}
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big] \\
&- \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x-w)^2}{\left[(x-z)^2-(x-w)^2\right]}\, \frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]} \\
&\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2} \\
&+ \frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\times \\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]\times \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big] \\
&- \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-w')^2}{\left[(x'-w')^2-(x'-z)^2\right]}\, \frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]} \\
&\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2} \\
&+ \frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\times \\
&\log\left[\frac{\frac{k^+}{\vee}\,(x'-w')^2+\left(1-\frac{k^+}{\vee}\right)(x'-z)^2}{(x'-w')^2+\frac{\Lambda}{k^+}\,(x'-z)^2}\right]\times \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big]
\end{aligned}
\tag{2.6}
$$

$$
\begin{aligned}
\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\, \frac{g^4\, d_\perp}{32\pi^5}\, \left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{\left[(x'-w')^2\,(x-w)^2-(x-z)^2\,(x'-z)^2\right]}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]\,\left[(x'-z)^2-(x'-w'\right.} \\
&\Big[f^{c'bd'}\, U^{ac'}(w')\, \rho^{d'}(x') - f^{ac'd'}\, U^{bc'}(z) U^{e'd'}(x')\, \rho^{e'}(x')\Big]\times\, \Big[f^{cbd}\, U^{ac}(w)\, \rho^{d}(x) - f^{acd}\, U^{bc}(z)\, U^{de}(x)\, \rho^{e}(x)\Big] \\
&- \frac{1}{(2\pi)^3}\, \frac{g^4\, d_\perp}{32\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x-z)^2\,(x-w)^2}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]}\, \frac{\left[(x'-z)^2-(x'-w')^2\right]}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]} \\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]\times \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big] \\
&+ \frac{1}{(2\pi)^3}\, \frac{g^4\, d_\perp}{32\pi^5}\, \left[\frac{1}{\vee}-\frac{1}{k^++\Lambda}\right] \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{1}{(z-w)^2\,(z-w')^2} \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big] \\
&+ \frac{1}{(2\pi)^3}\, \frac{g^4\, d_\perp}{32\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-w')^2\,(x'-z)^2}{(z-w)^2\,(z-w')^2}
\end{aligned}
$$

$$
\frac{\left[(x-z)^2-(x-w)^2\right]}{\left[(x'-z)^2-(x'-w')^2\right]\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}
$$
$$
\times\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{\frac{\Lambda}{k^+}\,(x'-z)^2+(x'-w')^2}\right]\times
$$
$$
\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$
$$
-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big] \tag{2.7}
$$

2.2 Row II

NA

2.3 Row III

NA

2.4 Row IV

$$
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]
$$
$$
\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}\right]+\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{\frac{1}{\xi}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}-e^{-ik\cdot(y'-y)}}{1-\xi}\right]
$$
$$
\times\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)
$$
$$
+\,\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.
$$
$$
\left.-\,\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)
$$
$$
+\,\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}}d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\right]
$$
$$
\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)
$$
$$
-\,\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}
$$
$$
\left[U^{e'c}(z)-U^{e'c}(x')\right]\times\left[U^{ec}(z)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x) \tag{2.8}
$$

Change C1. At $\xi\to\frac{k^+}{\vee}$ the phase $e^{-ik\cdot(y'-y)\frac{1}{\xi}}$ of line 1 probes $|y'-y|\sim\frac{k^+}{\vee|k|}$. The large-momentum tail of $\log(y'-y)^2$ then gives $\log\frac{\vee}{k^+}H_4$, with H_4 the bracket of the blue line ($y,y'\to z$: the measured gluon and its conjugate are emitted by the soft gluon z). $\log\frac{\vee}{k^+}H_4$ is one of the BFKL terms. The blue line takes it out here, so that it is not counted twice. Lines 1 and 2 with the blue line make up the DGLAP term; line 3 (linear polarization) is finite (`Finite_terms_final.pdf`, Sec. 4).

3 Group III (with c.c.)

3.1 Row I

$$
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'}e^{-ik(w'-w)}\left\{\right.
$$
$$
-\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2+\frac{\Lambda}{k^+}\,(x-z)^2}\right]\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.
$$
$$
\left.\left.-\,\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]\right.
$$
$$
+\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}
$$

$$
\begin{aligned}
&- \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times\\
&\left[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y)-U^{bd'}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\right]\times\\
&\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{3.1}
$$

3.2 Row II

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'}e^{-ik(w'-w)}\Bigg\{
&-\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2+\frac{\Lambda}{k^+}\,(x-z)^2}\right]\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.\\
&\left.-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}\right]\\
&+\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\\
&-\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times\\
&\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x')-U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{3.2}
$$

3.3 Row III

3.3.1 First part

$$
\begin{aligned}
\frac{d^3N_{1\,\rm NLO}}{d^3k}=&\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x}e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left[\frac{1}{p^+}\right]_+^{(k^+)}\delta_{jm}\,\delta_{ik'}-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{w,x',y',z,y,x}e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]\\
&+\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\,\frac{1}{p^++k^+}\int_{w,x',y',z,y,x}e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}\\
&\left[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]
\end{aligned}
\tag{3.3}
$$

Change C2. $\left[\frac{1}{p^+}\right]_+ \to \left[\frac{1}{p^+}\right]_+^{(k^+)}$ (kept unevaluated). $\int_0^{\vee-k^+}dp^+\left[\frac{1}{p^+}\right]_+F=\int_0^{\vee-k^+}dp^+\left[\frac{1}{p^+}\right]_+^{(k^+)}F-F(0)\log\frac{\vee}{k^+}$, and $-F(0)\log\frac{\vee}{k^+}$ is in the BFKL list. With $\left[\frac{1}{p^+}\right]_+$ it is counted twice. **Change C3.** The last (blue) line takes out the term (B) of this row (the region $y'\to z$), which is in the BFKL list (`BFKL_final.pdf`, terms (B)). After $\int_{y'}$, at large p^+, the $-\frac{1}{k^+}\delta_{im}\delta_{jk'}$ term equals minus the p^+ integrand of the blue line, up to terms that fall like $\frac{1}{p^{+2}}$ (Fourier lemma). Each alone grows like $\log\frac{\vee}{k^+}$; their sum is finite for $\vee\to\infty$. The p^+ integral of the blue line is $-$(term (B)).

3.3.2 Second part

$$
\begin{aligned}
\frac{d^3N_{2\,\rm NLO}}{d^3k}=&\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,w',y',z,y,x w,x',y',z,x}e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}}{(y'-z)^2\,(x'-y')^2}\,\frac{1}{\left[p^+\,(x-z)^2+k^+\,(x-w)^2\right]}\\
&\left[\frac{\delta_{k'm}\,d_\perp\,p^+}{2\,(z-w)^2\,(p^++k^+)^2}\,\left[(x-z)^2-(x-w)^2\right]+\frac{p^+\,\delta_{k'm}}{k^+\,(p^++k^+)}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2}-\frac{(x-z)\cdot(x-w)}{2\,(x-w)^2}\right.\right.
\end{aligned}
$$

$$
- \frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\Bigg] + \frac{\delta_{k'm}}{(p^+ + k^+)}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2} + \frac{(x-z)\cdot(x-w)}{2\,(x-z)^2} - \frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]
$$

$$
- \left[\frac{1}{p^+}\right]_+^{(k^+)}\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2} + \frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right] - \frac{p^+}{k^{+2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2} - \frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]
$$

$$
- \frac{1}{k^+}\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2} - \frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2} + \frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2} + \frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\Bigg]\times
$$

$$
\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
$$

$$
- \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{w,y',z,y,xw,x',y',z,x} e^{-ik\cdot(y'-w)}\left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right.
$$

$$
\left.+\frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right]\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times
$$

$$
\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
$$

$$
+ \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\,\frac{1}{k^+\,(p^++k^+)}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,\frac{(x'-z)\cdot(x-z)}{(x'-z)^2\,(x-z)^2}
$$

$$
\left[\frac{(y'-z)\cdot(z-w)}{(y'-z)^2\,(z-w)^2} - \frac{(y'-z)\cdot(x-w)}{2\,(y'-z)^2\,(x-w)^2}\right]\left[f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]
$$

$$
\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{3.4}
$$

Change C2. $\left[\frac{1}{p^+}\right]_+ \to \left[\frac{1}{p^+}\right]_+^{(k^+)}$ (kept unevaluated). $\int_0^{\vee-k^+} dp^+\left[\frac{1}{p^+}\right]_+ F = \int_0^{\vee-k^+} dp^+\left[\frac{1}{p^+}\right]_+^{(k^+)} F - F(0)\log\frac{\vee}{k^+}$, and $-F(0)\log\frac{\vee}{k^+}$ is in the BFKL list. With $\left[\frac{1}{p^+}\right]_+$ it is counted twice. Here $F(p^+) = \frac{e^{-ik\cdot(y'-z)p^+/k^+}}{p^+(x-z)^2+k^+(x-w)^2}$. **Change C3.** The last (blue) line takes out the term (B) of this row (the region $y'\to z$), which is in the BFKL list (`BFKL_final.pdf`, terms (B)). After $\int_{y'}$, at large p^+, the $-\frac{p^+}{k^{+2}}[\dots]$ term ($\to -\frac{1}{k^{+2}(x-z)^2}[\dots]$) equals minus the p^+ integrand of the blue line, up to terms that fall like $\frac{1}{p^{+2}}$ (Fourier lemma). Each alone grows like $\log\frac{\vee}{k^+}$; their sum is finite for $\vee\to\infty$. The p^+ integral of the blue line is $-$(term (B)). **Change C5.** Measure: the row has $\rho(x')\rho(x)$ and the kernels depend on w, x', y', z, x; there is no y and no w'.

3.4 Row IV

NA

3.5 Row V

$$
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}
$$

$$
\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)} - \left[\frac{1}{p^+}\right]_+^{(k^+)}\delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\right]\times
$$

$$
\left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right]
$$

$$
- \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}
$$

$$
\left[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right]
$$

$$
+ \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\,\frac{1}{p^++k^+}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}
$$

$$
\left[U^{e'c'}(z)\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right] \tag{3.5}
$$

Change C2. $\left[\frac{1}{p^+}\right]_+ \to \left[\frac{1}{p^+}\right]_+^{(k^+)}$ (kept unevaluated). $\int_0^{\vee-k^+} dp^+\left[\frac{1}{p^+}\right]_+ F = \int_0^{\vee-k^+} dp^+\left[\frac{1}{p^+}\right]_+^{(k^+)} F - F(0)\log\frac{\vee}{k^+}$, and $-F(0)\log\frac{\vee}{k^+}$ is in the BFKL list. With $\left[\frac{1}{p^+}\right]_+$ it is counted twice. **Change C3.** The last (blue) line takes out the term (B) of this row (the region $y'\to z$), which is in the BFKL list (`BFKL_final.pdf`, terms (B)). After $\int_{y'}$, at large p^+, the $-\frac{1}{k^+}\delta_{im}\delta_{jk'}$ term equals minus the p^+ integrand of the blue line, up to terms that fall like $\frac{1}{p^{+2}}$ (Fourier lemma). Each alone grows like $\log\frac{\vee}{k^+}$; their sum is finite for $\vee\to\infty$. The p^+ integral of the blue line is $-$(term (B)).

3.6 Row VI

$$
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}
$$

$$\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \left[\frac{1}{p^+}\right]_+^{(k^+)} \delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right] \Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big] \times$$
$$\big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]$$
$$- \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1 - \frac{k^+}{\vee}\right] \int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big] \times \big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]$$
$$+ \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+} \int_0^{\vee - k^+} \frac{dp^+}{2\pi}\,\frac{1}{p^+ + k^+} \int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big] \times \big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big] \qquad (3.6)$$

Change C2. $\left[\frac{1}{p^+}\right]_+ \to \left[\frac{1}{p^+}\right]_+^{(k^+)}$ (kept unevaluated). $\int_0^{\vee - k^+} dp^+ \left[\frac{1}{p^+}\right]_+ F = \int_0^{\vee - k^+} dp^+ \left[\frac{1}{p^+}\right]_+^{(k^+)} F - F(0)\log\frac{\vee}{k^+}$, and $-F(0)\log\frac{\vee}{k^+}$ is in the BFKL list. With $\left[\frac{1}{p^+}\right]_+$ it is counted twice. **Change C3.** The last (blue) line takes out the term (B) of this row (the region $y' \to z$), which is in the BFKL list (`BFKL_final.pdf`, terms (B)). After $\int_{y'}$, at large p^+, the $-\frac{1}{k^+}\delta_{im}\delta_{jk'}$ term equals minus the p^+ integrand of the blue line, up to terms that fall like $\frac{1}{p^{+2}}$ (Fourier lemma). Each alone grows like $\log\frac{\vee}{k^+}$; their sum is finite for $\vee \to \infty$. The p^+ integral of the blue line is $-$(term (B)).